# Seeing Motion in Data: Applying Image Formation Theory to Human Activity Recognition

A Deep Learning Exploration Using Inertial Sensor Signals

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**Abstract**

What if you could take a stream of numbers from a smartphone's sensors and literally *paint* it into a picture that a neural network could classify? That is the core idea behind the work of Liu et al. [1], and it is what this project explores.

We implement their **image formation theory** on the UCI Human Activity Recognition (HAR) dataset [2], which records nine inertial channels from 30 volunteers performing everyday activities. The theory provides a pipeline of three operators—$\alpha$, $\beta$, and $\gamma$—that transforms raw 1-D time-series into $24 \times 48$ grayscale images.

We run **24 experiments**: four CNN architectures (LeNet-5, VGG-Style, ResNet-Style, Lightweight) crossed with six $\gamma$ operators. Our best result—**92.13% accuracy** (F1-macro **92.12%**)—is obtained by ResNet-Style paired with $\gamma_3$ (Centered Body Acceleration). We find that model architecture matters considerably more than signal arrangement ($\sim$5.4 vs. $\sim$2.6 percentage-point effect), that no single $\gamma$ operator is universally best, and that the Sitting/Standing confusion is a fundamental sensor-physics bottleneck.

**Keywords:** image formation, deep learning, time-series classification, human activity recognition, convolutional neural networks, sensor fusion

# 1 Motivation and Problem Setting

Modern smartphones carry accelerometers and gyroscopes that record motion along three orthogonal axes at 50 samples per second. An activity-recognition system must translate these nine streams of numbers into labels like *walking*, *sitting*, or *lying down*.

Classifying these signals directly is deceptively difficult: raw acceleration for "walking upstairs" can look similar to "walking on flat ground", magnitudes drift between subjects, and temporal dependencies span dozens of samples. Traditional approaches resort to hand-crafted features, but these require domain expertise and generalise poorly.

Convolutional Neural Networks (CNNs) excel at images—picking up spatial patterns, textures, and hierarchical structures via local, shift-invariant kernels. If we could *cast our time-series problem as an image problem*, we could inherit decades of CNN progress for free.

Liu et al. [1], published in *IEEE Transactions on Industrial Informatics* (2021), provide a mathematically rigorous answer: **image formation theory**. They demonstrated 96.09% accuracy on a photovoltaic fault-detection dataset with 23 sensors using LeNet-5 on the resulting images.

Our project asks: *does this theory work on human activity recognition*—a physically different domain with fewer channels (9 vs. 23)? We extend the paper by comparing **four CNN architectures** across **six signal-arrangement operators**.

# 2 Theoretical Framework

The image formation pipeline applies three operators in sequence: $\alpha \to \beta \to \gamma$. In short, $\alpha$ sorts the signals, $\beta$ normalises their values, and $\gamma$ positions them spatially inside an image frame.

## 2.1 The $\alpha$ Operator: Signal Ordering

The $\alpha$ operator is an invertible permutation that gathers independently collected signals into a prescribed sequence: $\mathbf{e}_i^T = \alpha\, \mathbf{e}_j^T$. For the HAR dataset we index the nine channels as:

- Channels 0–2: body acceleration ($x$, $y$, $z$), gravity removed
- Channels 3–5: body gyroscope ($x$, $y$, $z$), angular velocity
- Channels 6–8: total acceleration ($x$, $y$, $z$), raw accelerometer

### 2.2 The $\beta$ Operator: Normalisation

All channels must share the same amplitude scale before becoming pixel values. The $\beta$ operator applies channel-wise min-max normalisation:

$$\hat{x}_i(k) = \frac{x_i^{\text{raw}}(k) - \min_a x_i^{\text{raw}}(a)}{\max_b x_i^{\text{raw}}(b) - \min_a x_i^{\text{raw}}(a)} \tag{1}$$

This maps every signal to $[0, 1]$ ($0 =$ black pixel, $1 =$ white). We fit the min/max on the 7,352 training windows only and apply the same parameters to the test set, preventing information leakage. A division-by-zero guard handles any constant-valued channels.

### 2.3 The $\gamma$ Operator: Image Formation

At each time step $k$, the $\gamma$ operator maps 9 normalised values to a $3 \times 3$ pixel patch ($h = l = 3$, no padding needed):

$$\gamma(k) : \mathbf{x}(k) \mapsto \mathbf{y}(k) \in [0, 1]^{3\times 3} \tag{2}$$

The *arrangement* of signals within the patch is what distinguishes $\gamma_1$ through $\gamma_6$—the creative heart of the framework.

### 2.4 Sequential Image Formation and Rigidity

A full 2.56-second window has 128 time steps, producing 128 patches. The sequential operator tiles these into an $8 \times 16$ grid, giving a final image of $(8 \times 3) \times (16 \times 3) = 24 \times 48$ pixels, where the horizontal axis encodes time and the vertical axis encodes sensor arrangement.

Liu et al. prove (Lemma 2) that the composed mapping $\tilde{\gamma} = \gamma \circ \beta \circ \alpha$ is **rigid**: the same raw signal always produces the same image—a deterministic, lossless transformation that lets a CNN generalise predictably.

## 3 Experimental Setup

### 3.1 The UCI HAR Dataset

Collected by Anguita et al. [2] from 30 volunteers (aged 19–48) wearing a Samsung Galaxy S II on their waist, the dataset records six activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, and LAYING. The accelerometer and gyroscope were sampled at 50 Hz, with 2.56-second sliding windows (128 samples, 50% overlap). Train/test split: **7,352 / 2,947** windows, partitioned at the *subject* level for person-

independent evaluation. The class distribution is nearly balanced (13–18% per activity), so accuracy and F1-macro are essentially interchangeable metrics.

Figure 1: Raw 9-channel sensor signals for one representative window of each activity. LAYING is clearly distinct; SITTING and STANDING are nearly indistinguishable.

### 3.2 The Six $\gamma$ Operators

We designed six spatial arrangements reflecting different hypotheses about which signals should be adjacent (Table 1, Fig. 2):

- $\gamma_1$ **(Component-wise)**: each row = one sensor type. Most physically interpretable.
- $\gamma_2$ **(Accel vs Gyro)**: translational sensors clustered together, rotational separate.
- $\gamma_3$ **(Centered Body Acc)**: body acceleration in the centre row, flanked by gyroscope and total acceleration. Mirrors Liu et al.'s centred-current arrangement.
- $\gamma_4$ **(Centered Gyro)**: gyroscope centred, acceleration surrounding.
- $\gamma_5$ **(Cross-modal mix 1)**: one channel from each sensor type per row, maximising

cross-modal interaction within a kernel's receptive field.

- $\gamma_6$ **(Cross-modal mix 2)**: another cross-modal permutation for robustness checking.

Table 1: The six $\gamma$ operators. Each cell shows the signal index placed at that pixel location in the $3 \times 3$ per-step patch.

| Op. | Design Principle | Arrangement |
|---|---|---|
| $\gamma_1$ | Component-wise | $\begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix}$ |
| $\gamma_2$ | Accel vs. Gyro | $\begin{pmatrix} 0 & 1 & 2 \\ 6 & 7 & 8 \\ 3 & 4 & 5 \end{pmatrix}$ |
| $\gamma_3$ | Body acc centred | $\begin{pmatrix} 3 & 6 & 4 \\ 0 & 1 & 2 \\ 5 & 7 & 8 \end{pmatrix}$ |
| $\gamma_4$ | Gyro centred | $\begin{pmatrix} 0 & 6 & 1 \\ 3 & 4 & 5 \\ 2 & 7 & 8 \end{pmatrix}$ |
| $\gamma_5$ | Cross-modal mix 1 | $\begin{pmatrix} 6 & 0 & 4 \\ 2 & 7 & 3 \\ 8 & 1 & 5 \end{pmatrix}$ |
| $\gamma_6$ | Cross-modal mix 2 | $\begin{pmatrix} 5 & 7 & 0 \\ 3 & 8 & 1 \\ 6 & 2 & 4 \end{pmatrix}$ |

### 3.3 The Four CNN Architectures

We compared four architectures spanning a wide range of complexity to separate the effect of the image representation from the classifier on top:

Table 2: CNN architectures. All accept $24 \times 48 \times 1$ inputs and output 6-class softmax.

| Architecture | Params | Key Design |
|---|---|---|
| LeNet-5 (adapted) | 609,542 | Two conv layers (32, 64), dense head [6] |
| VGG-Style | 763,878 | Stacked $3 \times 3$ convs + BatchNorm [3] |
| ResNet-Style | 326,086 | 3 residual blocks (32, 64, 128) + GAP [4] |
| Lightweight | 39,334 | Depthwise separable convs (MobileNet-style) [5] |

All models use Adam (lr $= 10^{-3}$), reduce-on-plateau (factor 0.5, patience 4), and early stopping (patience 8, restoring best weights). Seed = 42; TensorFlow 2.19.0.

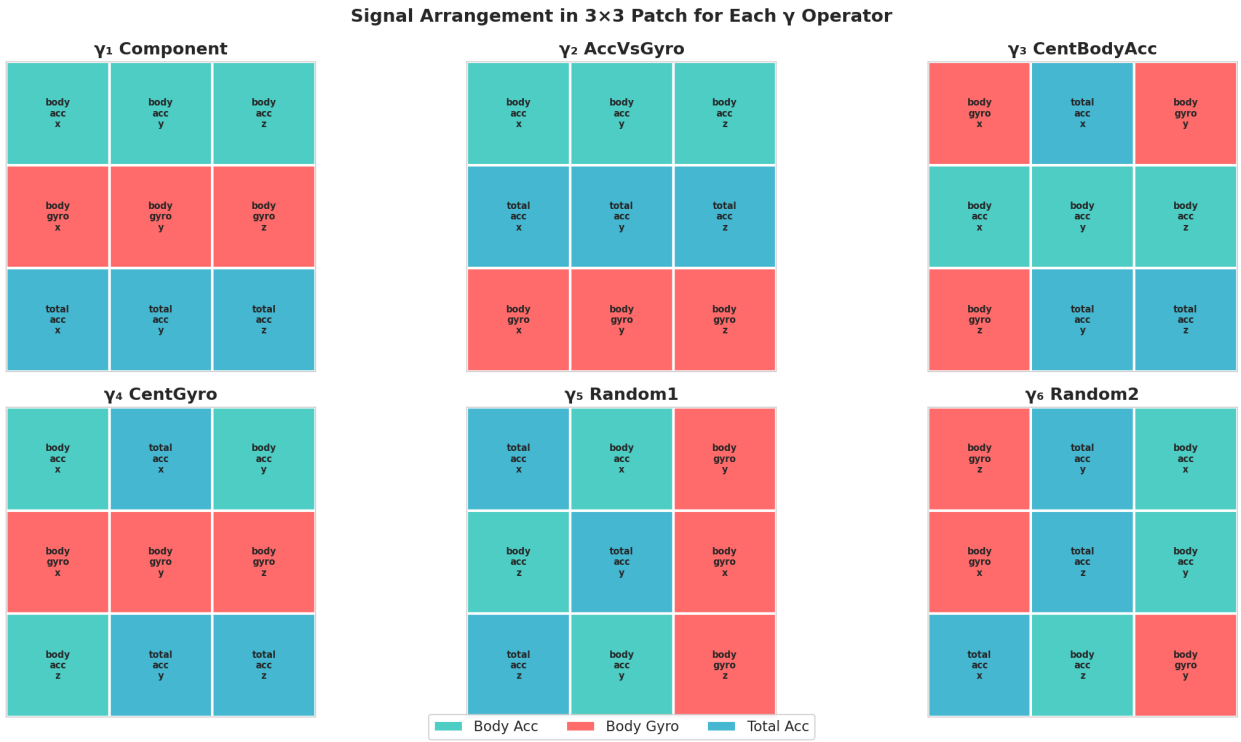


Figure 2: Visual comparison of signal-to-pixel mapping for each $\gamma$ operator.

# 4 Results

## 4.1 24 Experiments at a Glance

Table 3 presents test-set accuracy for every model–operator combination.

Table 3: Test accuracy (%) for 4 models × 6 $\gamma$ operators. Bold = best per row.

| **Model** | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ | $\gamma_4$ | $\gamma_5$ | $\gamma_6$ | **Best** | **Avg** |
|---|---|---|---|---|---|---|---|---|
| LeNet-5 | 80.0 | 86.8 | 83.4 | **88.5** | 87.1 | 87.9 | 88.5 | 85.6 |
| VGG-Style | 89.6 | 87.8 | 89.8 | 85.4 | 89.4 | **91.3** | 91.3 | 88.9 |
| ResNet-Style | 90.7 | 90.7 | **92.1** | 91.2 | 89.7 | 91.7 | **92.1** | 91.0 |
| Lightweight | 90.1 | 90.2 | 89.4 | 90.7 | **91.7** | 89.9 | 91.7 | 90.3 |
| **Col avg** | 87.6 | 88.9 | 88.7 | 89.0 | 89.5 | 90.2 | — | — |

The overall winner is **ResNet-Style + $\gamma_3$: 92.13% accuracy, F1-macro = 92.12%**. Fig. 4 visualises all 24 results.

## 4.2 Per-Class Behaviour

The confusion matrices (Fig. 5) and per-class heatmap (Fig. 6) reveal a consistent pattern:

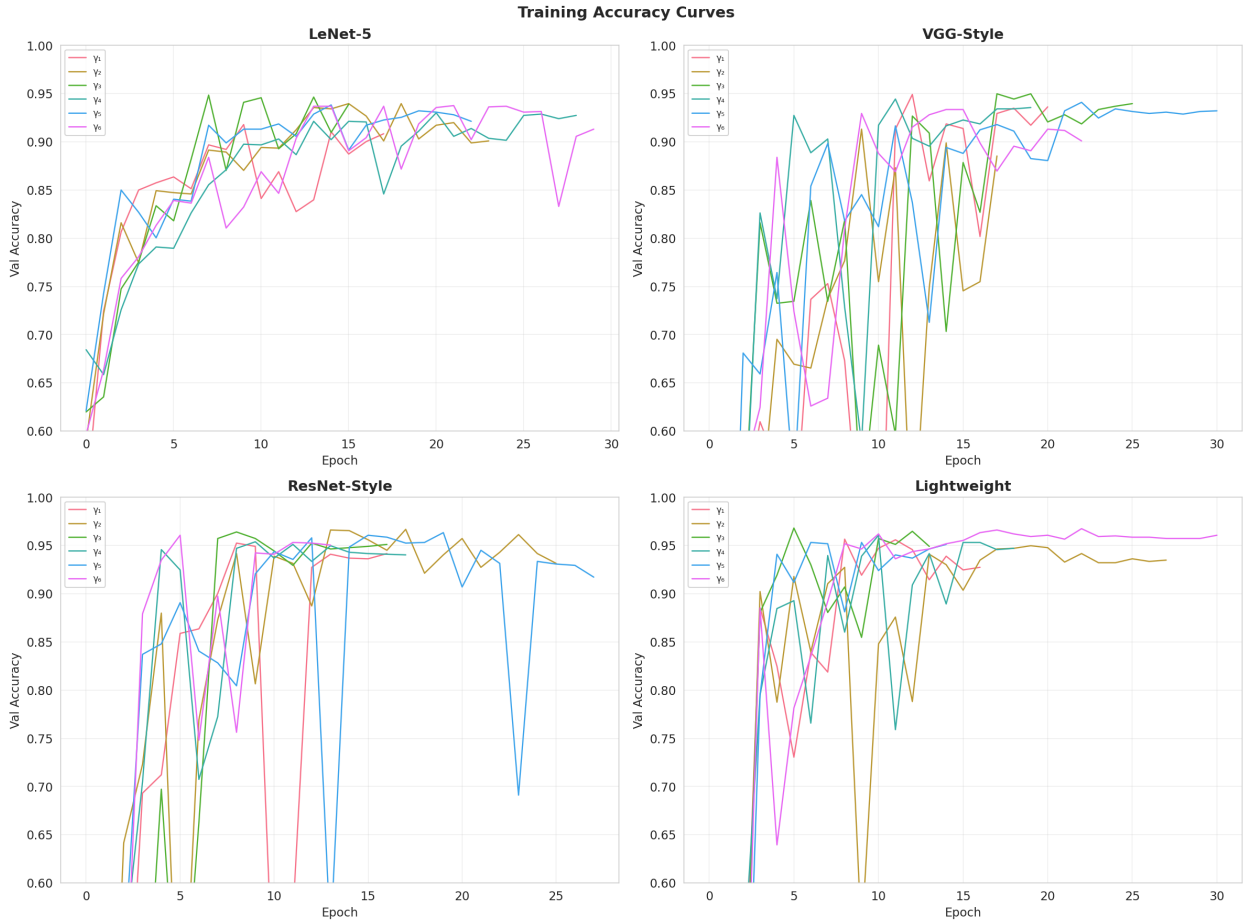


Figure 3: Training and validation accuracy curves for all 24 experiments. Most models converge within 30–50 epochs.

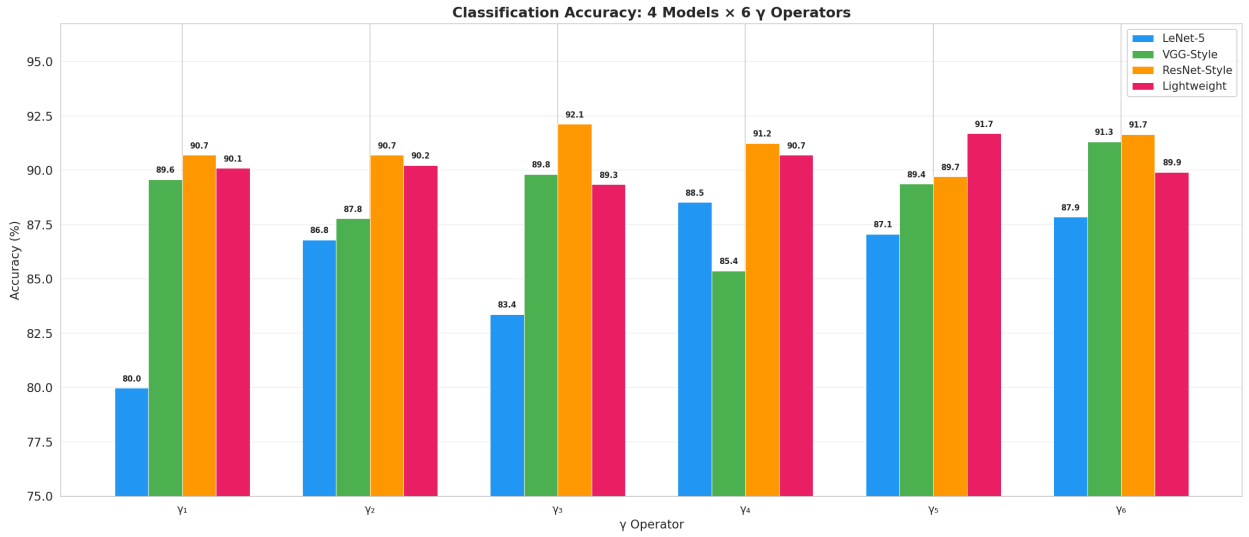


Figure 4: Accuracy across all models and $\gamma$ operators. ResNet sits consistently near the top; LeNet-5 is the most sensitive to operator choice (8.5pp range).

- **LAYING: 100% accuracy** in every configuration. When lying down, gravity acts fully along a horizontal body axis, producing a unique accelerometer signature.
- **Walking variants: 87–98%**. The gait cycle for upstairs/downstairs walking shares the same fundamental frequency as level walking, differing mainly in vertical acceleration amplitude. The image representation captures this as a texture difference, but not perfectly.
- **SITTING/STANDING: 77–88%**—the project's Achilles heel. Both produce near-identical signals: the body is stationary, gravity acts along the same axis, and the only difference is a slight shift in trunk orientation. This is a genuine sensor-physics limitation that no arrangement or architecture can fully resolve.

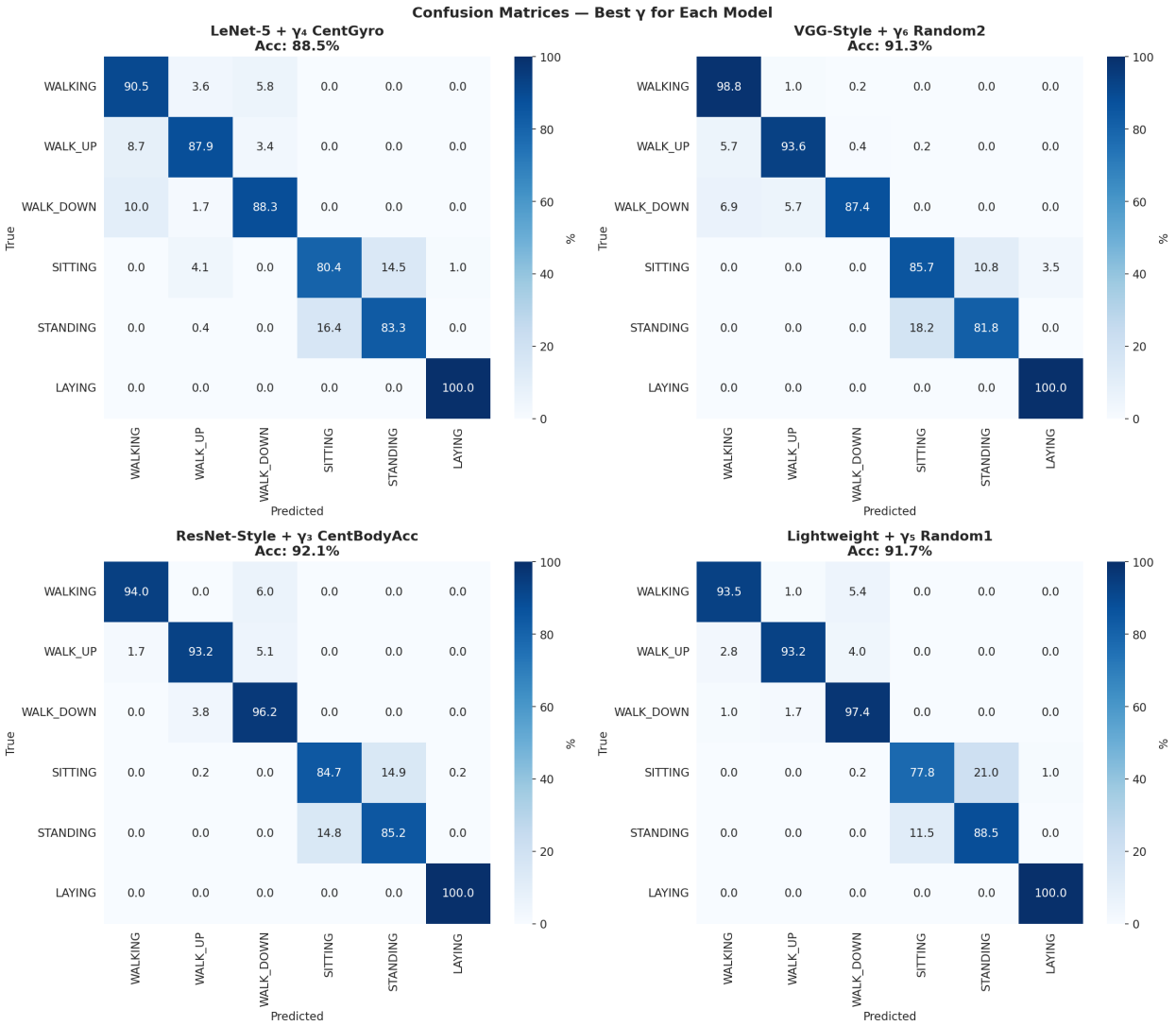


Figure 5: Confusion matrices for the best $\gamma$ of each model. The SITTING/STANDING block is visibly imperfect across all architectures.

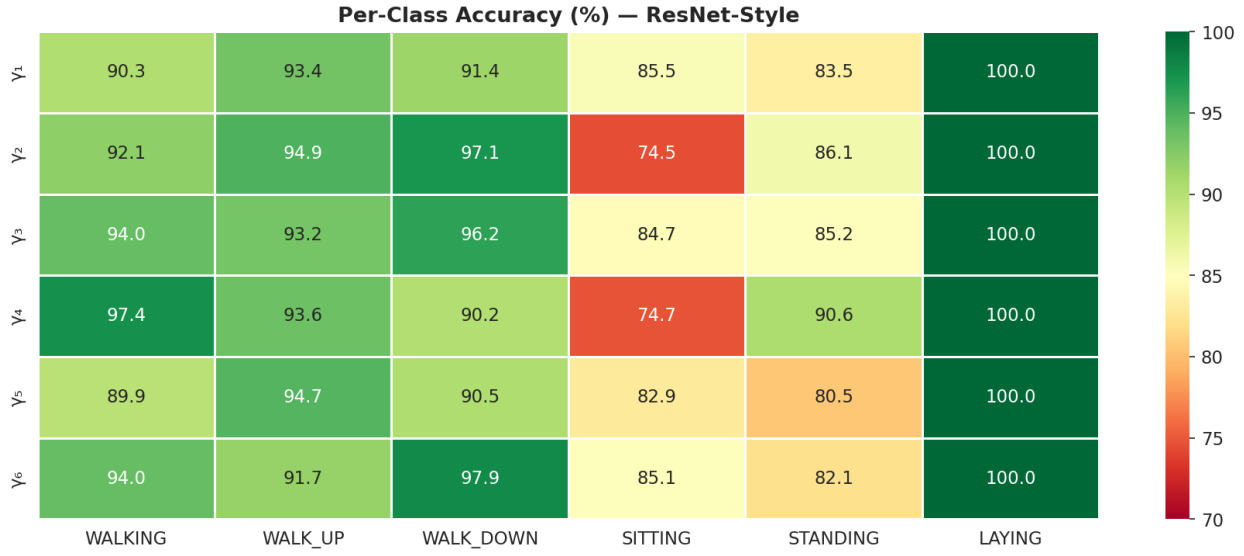


Figure 6: Per-class accuracy (%) for ResNet-Style across all six $\gamma$ operators.

### 4.3 Model Complexity vs. Accuracy

Fig. 7 plots best accuracy against parameter count. ResNet achieves the highest accuracy at 326K parameters. VGG uses 764K parameters yet falls short—the $24 \times 48$ images are too small for deep VGG-style representations to pay off. Remarkably, the **Lightweight model (39K params) achieves 91.7%**—within 0.44pp of the best, and roughly 8× smaller than ResNet and 20× smaller than VGG. This suggests image-formed representations are efficient enough for edge-deployable classifiers.

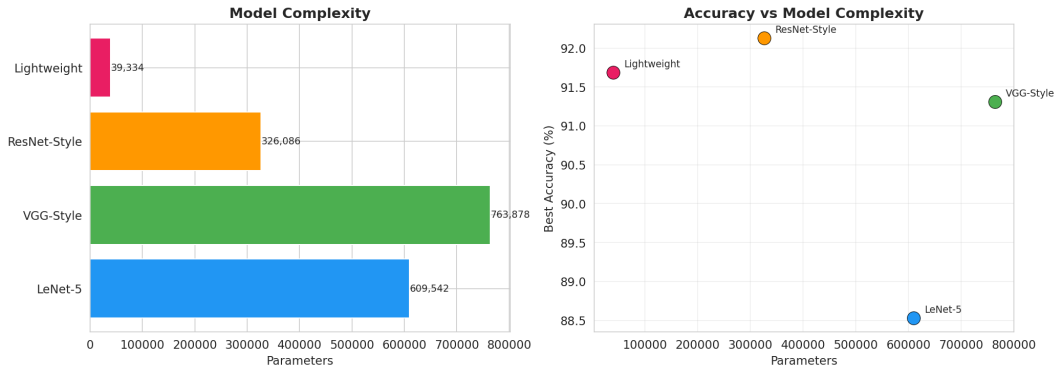


Figure 7: Best accuracy vs. parameter count. ResNet sits in an attractive sweet spot.

## 5 Analysis and Discussion

### 5.1 How Much Does $\gamma$ Actually Matter?

Column averages in Table 3 range from 87.6% ($\gamma_1$) to 90.2% ($\gamma_6$): a spread of **2.6pp**. Row averages range from 85.6% (LeNet-5) to 91.0% (ResNet): a spread of **5.4pp**. Model choice therefore matters *considerably* more than signal arrangement—roughly twice the effect size—but both effects are real and measurable.

### 5.2 Physically Motivated vs. Cross-Modal Mixing

The best $\gamma$ varies by model: LeNet-5 peaks at $\gamma_4$ (88.5%), VGG at $\gamma_6$ (91.3%), ResNet at $\gamma_3$ (92.1%), Lightweight at $\gamma_5$ (91.7%). No single operator dominates—*the optimal arrangement is architecture-dependent.*

Why does $\gamma_3$ work best for ResNet? Body acceleration (gravity-removed) is the cleanest activity-discriminative signal. Placing it in the centre row means every $3 \times 3$ kernel overlaps with at least one body-acceleration pixel. ResNet's skip connections then propagate gradients efficiently from the classifier head back to those first-layer filters.

By contrast, VGG and Lightweight benefit from cross-modal mixing ($\gamma_5/\gamma_6$). Without skip connections, their gradient flow is sequential. When different modalities are interleaved spatially, first-layer kernels can detect inter-modal correlations directly (e.g. "acceleration spiked *and* angular velocity spiked at this time step").

### 5.3 Why "Random" Arrangements Are Competitive

$\gamma_5$ and $\gamma_6$ frequently outperform the physically-grouped $\gamma_1$ and $\gamma_2$. The column average for $\gamma_6$ (90.2%) is the highest of all six. Activity recognition is fundamentally an *inter-sensor* problem: what makes walking distinctive is not just the accelerometer pattern but its coordination with the gyroscope pattern. Cross-modal arrangements force first-layer kernels to see both signals simultaneously within their receptive field.

This contrasts with Liu et al.'s finding that "randomly arranged patterns are prone to generate worse results" on their photovoltaic data. The difference likely stems from HAR having fewer, more tightly correlated sensor groups than the 23-channel photovoltaic setting.

### 5.4 Comparison with the Original Paper

Liu et al. report **96.09%** with LeNet-5 on their photovoltaic dataset. Our best is **92.13%** with ResNet on HAR. The gap is expected: (1) the photovoltaic dataset has 23 signals (richer images); (2) it classifies 3 conditions vs. our 6 activities; (3) the Sitting/Standing confusion is

a fundamentally hard problem for inertial sensors. Our 92.1% also exceeds the 88.5% of our LeNet-5, demonstrating that architectural innovation atop the image-formation framework yields gains the original paper did not explore.

## 6 Sample Images

Fig. 8 shows formed images for each activity across $\gamma$ operators. Dynamic activities show periodic horizontal banding from the gait cycle; LAYING produces near-constant heatmaps reflecting the static gravity vector; SITTING and STANDING look eerily similar, explaining the persistent classification difficulty. Cross-modal operators ($\gamma_5$, $\gamma_6$) produce distinctive diagonal textures from beating patterns between differently-oscillating sensor channels.

## 7 Avenues for Improvement

1. **Data augmentation**: Time warping, jitter, and magnitude scaling applied *before* image formation to increase training diversity.
2. **Multi-channel (RGB) images**: Use three different $\gamma$ operators as the R, G, B channels, converting a design choice into an enrichment.
3. **Frequency features**: Augment signals with FFT magnitudes, increasing $m$ for richer images.
4. **Learnable $\gamma$**: Replace the fixed permutation with a differentiable layer trained end-to-end—a soft attention mechanism over sensor positions.
5. **Tiling geometry**: Varying the grid (e.g. $16 \times 8$ or $4 \times 32$) can improve results by up to $\sim$5% for certain architectures.
6. **Hierarchical classification**: Separate dynamic vs. static activities first, then sub-classify to reduce Sitting/Standing and cross-walking confusions independently.

## 8 Conclusion

By implementing the image formation theory of Liu et al. [1] and extending it with four CNN architectures across six $\gamma$ operators (24 experiments), we map the design space for HAR. The key findings:

1. **92.13% accuracy** (ResNet + $\gamma_3$, F1 = 92.12%) demonstrates successful cross-domain transfer from photovoltaic to activity recognition.
2. **Architecture matters more than arrangement** ($\sim$5.4 vs. $\sim$2.6pp), but both effects

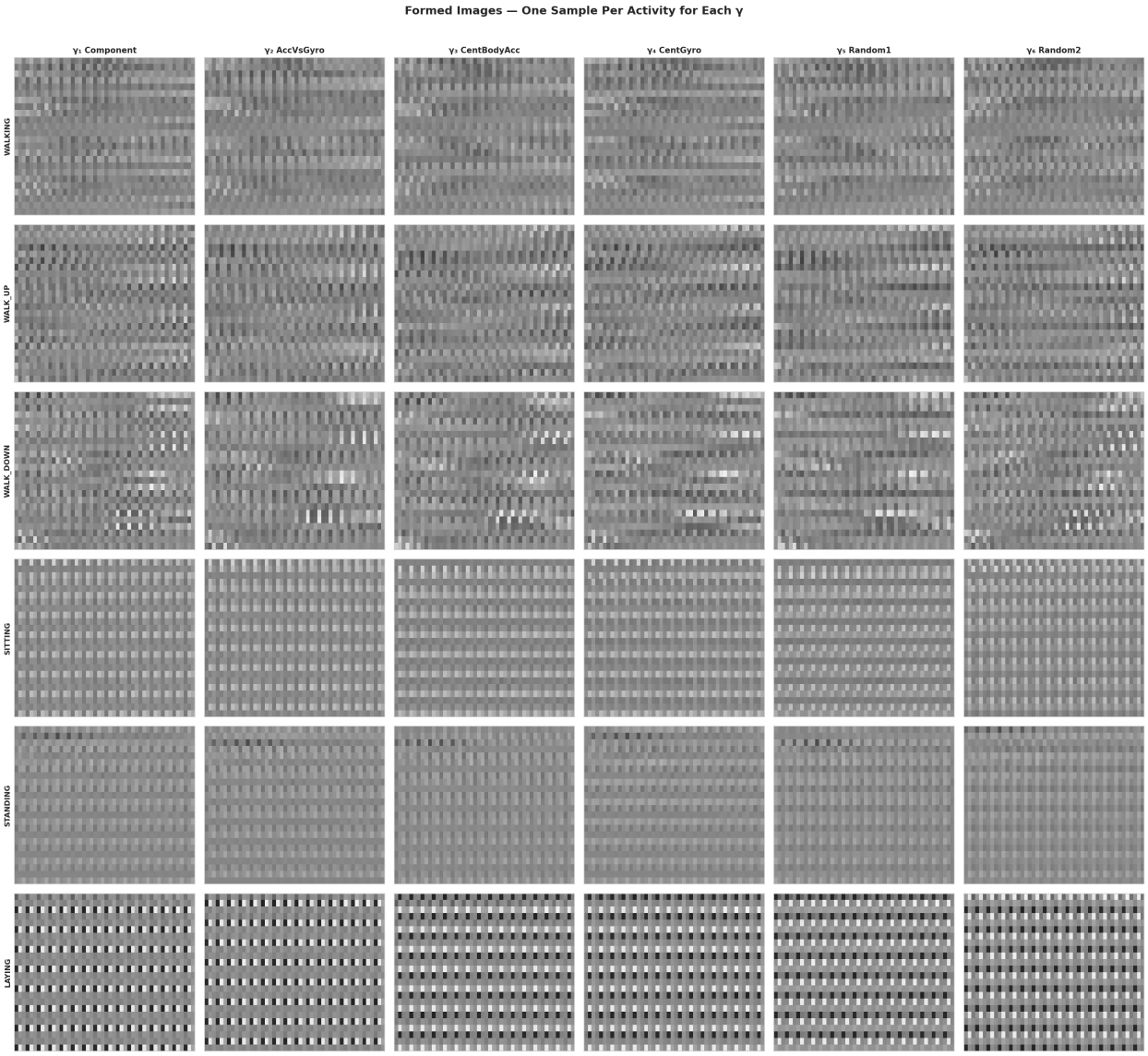


Figure 8: Formed $24 \times 48$ images per activity for each $\gamma$ operator.

are real and non-negligible.

3. **No $\gamma$ is universally best;** the optimal arrangement is architecture-specific. ResNet peaks at the physically-motivated $\gamma_3$; VGG and Lightweight peak at cross-modal mixing operators.
4. **Cross-modal mixing is competitive** despite appearing "random," because inter-sensor correlations are as discriminative as within-sensor patterns for HAR.
5. **The Lightweight model (39K params) achieves 91.7%**—within 0.44% of the best—suggesting image-formed representations are efficient enough for edge deployment.
6. **SITTING/STANDING confusion is the fundamental bottleneck,** driven by sensor physics rather than model capacity.

The image formation approach is elegant in its generality: it requires no hand-crafted features and no domain-specific preprocessing. Any dataset of multiple synchronised time-series—from industrial vibration monitoring to EEG brain-computer interfaces—could be analysed with the same pipeline.

## 8 References


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